

Naming Guidelines for Files, Modules, Products

Python Naming Rules

Element	Naming Rule	Example
Python files	lowercase with underscores for readability	vibm_scope_core.py
Folders/Packages	lowercase, no spaces; use underscores if needed	core, signal_processing
Product mains	PascalCase + Version suffix if needed	VibMScopeV1.py
Classes	PascalCase	SignalHandler, Logger
Functions/Methods	snake_case	validate_input(), run_loop()
Constants	UPPERCASE_WITH_UNDERSCORES	DEFAULT_SAMPLING_RATE
Config files	short + dashes or underscores	config_dev.json, paths.ini
Temp/test scripts	prefix with <i>test</i> or <i>dev</i>	_test_adc_driver.py
Versioned builds	productName + version	ProductionToolV1.3.exe
Firmware filenames	Include hardware version & build date optionally	fw_v2.1.05_20250610.hex

Auto-Versioning Strategy

Use a central file (e.g., version.py):

```
__version__ = "1.3.0"  
__build__   = "20250610"  
__product__ = "VibMScope"
```

Use this in: - GUI window titles - Log headers - File names (log_VibMScope_v1.3_20250610.txt) - PDF/export metadata - PyInstaller spec (--name)

Git + IDE + Tooling for Private Products

Private Git Hosting Options

Option	Notes
GitHub	Free private repos, easy to start with

Option	Notes
GitLab	Free with powerful CI/CD
Bitbucket	Integrates well with Atlassian tools

Recommendation: Use GitHub if you already have an account.

Free IDE Options

Use Case	IDE / Tool	Notes
Python GUI	VS Code or PyCharm CE	VS Code is light, PyCharm is full IDE
Firmware	STM32CubeIDE or Keil	STM32CubeIDE is more Git-friendly
MATLAB	MATLAB IDE	Use Git via system terminal
Hardware	CAD tools + Git folder	Save BOMs, layouts, configs

Recommended Git Repo Structure

```

my_project/
├── python/
│   ├── VibMScopeV1.py
│   ├── core/
│   └── utils/
├── firmware/
│   ├── STM32F4_VibMScope/
│   └── inc/, src/
├── matlab/
│   └── analysis_tools/
├── docs/
│   └── block_diagram.png, version_history.md
├── hardware/
│   └── schematics/, layout/
├── version.py
└── README.md

```

Optional Enhancements

- Auto-version stamping in logs and builds
- Per-product .gitignore and build scripts
- Selective GitHub publishing (partial folder push)

Final Verdict

Your current plan is modular, scalable, and version-safe: - Use clear names - Centralize versioning - Git-based control across all domains - Build per product easily

Let me know if you want a starter template, `.gitignore`, or build script.